



Analyse A Serie 4

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Génie Mécanique

1 Exercice 1

1.1 A

$$(m-2)x - (11m+3)x = -10$$

$$x(-10m-5) = -10$$

$$1. -10m-5 \neq 0 \Leftrightarrow m \neq \frac{1}{2} \quad S = \left\{ \frac{10}{10m+5} = \frac{2}{2m+1} \right\}$$

$$2. S = \emptyset$$

1.2 B

$$1. m \neq -\frac{1}{2} \quad S = 2m-1$$

$$2. m = \frac{1}{2} \quad S = \mathbb{R}$$

2 Exercice 8

$$x^2 + (m-2)x - (m+3) = 0$$

$$\Delta = (m-2)^2 - 4(-m-3)$$

$$x_1 = \frac{-m+2 + (m-2)^2 - 4(-m-3)}{2}$$

$$x_2 = \frac{-m+2 - ((m-2)^2 - 4(-m-3))}{2}$$

$$\left(\frac{-m+2 + (m-2)^2 - 4(-m-3)}{2} \right)^2 + \left(\frac{-m+2 - ((m-2)^2 - 4(-m-3))}{2} \right)^2 = 9$$

$$(-m+m^2+9)^2 + (-1-m^2+7m)^2 = 36$$

$$(m(m-1)+9)^2 + (m(m+7)-1)^2 = 36$$

$$(m(m-1))^2 + 18m(m-1) + 81 + (m(m+7))^2 - 2m(m+7) + 1 = 36$$

$$m^2(m^2-2m+1) + 18m^2 - 18m + 82 + m^2(m^2+14m+49) - 2m^2 + 14m = 36$$

$$2m^4 + 12m^3 + 66m^2 - 4m + 82 = 36$$

Autre Chemin Formule de viète $x^2 + (m-2)x - (m+3) = 0$

$$x_1 + x_2 = -\frac{a_2}{a_1}$$

$$x_1 + x_2 = -\frac{m-2}{1}$$

$$x_1 + x_2 = -m+2$$

$$(x_1 + x_2)^2 = x_1^2 + x_2^2 + 2x_1 * x_2$$

$$(-m + 2)^2 = 9 + 2x_1x_2$$

$$x_1x_2 = \frac{a_0}{a_2}$$

$$x_1x_2 = (-m - 3)$$

$$9 = (-m + 2)^2 + 2(m + 3) = m^2 - 4m + 4 + 2m + 6 = m^2 - 2m + 10$$

$$\Delta = 4 - 4 = 0$$

$$x_1 = 2\frac{m}{2} = m$$

$$(m - 1)^2 = 0$$

$$m = 1$$